

Parth Joshi

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Education

Indian Institute of Technology, Mandi

B.Tech in Data Science

Dec 2020 – May 2024

Technical Skills

LLM & GenAI Systems: LangChain, LangGraph, LangExtract, MCP (Model Context Protocol), llama.cpp, GGUF Quantization, Multi-Agent Workflows, Context Management, SillyTavern, Prompt Engineering, RAG Architectures, LLM Evaluation

Agentic AI & Production Systems: Multi-Agent Orchestration, Tool-Use Integration, Autonomous Reasoning Pipelines, State-Driven Dialogue Management, Model Serving, ML Infrastructure, MLOps

NLP & Deep Learning: BERT, InLegalBERT, HuggingFace Transformers, Embedding Fine-tuning, Intent Classification, Semantic Search, Named Entity Recognition

ML Engineering: TensorFlow, PyTorch, Keras, Scikit-Learn, LSTM, OpenCV, Feature Engineering, Model Optimization

Data Engineering: PostgreSQL, Redis, Kafka, Docker, Kubernetes, ETL Pipelines, Real-time Processing, Pinecone, ChromaDB, REST APIs

Programming & Tools: Python, SQL, C++, JavaScript, Django, Flask, FastAPI, Streamlit, Git, Pandas, NumPy, Matplotlib, Plotly, Playwright, BeautifulSoup, Locust

Experience

Data Scientist

Oct 2025 – Present

Amenify, Gurgaon

- Architected Maddie AI, a production multi-agent conversational AI system using LangGraph and LangChain with multi-agent orchestration and deterministic state-driven dialogue management, serving 10K+ monthly bookings at scale
- Developed autonomous reasoning pipelines with tool-use integration for automated home service workflows, demonstrating production-grade LLM system deployment with inference optimization
- Built location-based restaurant data service using Playwright and BeautifulSoup to bypass bot detection, managing headless browser sessions and exposing real-time menu data via REST API endpoints; validated scalability with Locust load testing
- Write complex SQL queries for Trevor client dashboards to surface performance metrics, operational KPIs, and business intelligence insights

Software Developer

Aug 2024 – Mar 2025

Samarth Egov, Ministry of Education, New Delhi

- Developed the Samarth E-Gov Suite using Yii and Bootstrap, enabling 200+ universities to streamline administration
- Optimized workflows, reducing system response time by 30% and improving operational efficiency for over 900,000 students and faculty

Data Analyst Intern

Feb 2023 – July 2023

Axetrack Software Solutions, Jaipur

- Cross-validated transition from scheduler-based to push-based data processing using Kafka, reducing data ingestion latency for real-time logistics tracking
- Improved automated OCR pipeline for document categorization and routing, processing 10,000+ files with enhanced accuracy
- Revamped ETL pipeline architecture to optimize logistics data handling, achieving 30% increase in data retrieval speed

Research

Predicting Magnetization of Transition Metal-Based Materials Using CGCNN

Sep 2025 [↗](#)

Published in *Journal of Magnetism and Magnetic Materials* (Elsevier). Developed Crystal Graph Convolutional Neural Networks to predict magnetization in transition metal compounds, reducing reliance on computationally expensive DFT calculations.

NLP & GenAI Engineering Projects

Dynamic Credit System (Geo-Aware Credit Intelligence)

In Progress

- Building rule-based dynamic credit limit engine using real-time location context and behavioral risk signals including spend velocity and utilization patterns
- Implemented geo-fencing with OpenStreetMap POIs for high-risk zone detection and triggered credit limit updates on geo-fence entry events
- Designed ML-ready architecture for future model-driven credit optimization and A/B testing framework

Email Intent Classification System

[GitHub](#) 

- Developed multi-class NLP pipeline for real estate email routing using BERT-based fine-tuning for intent classification with prompt engineering optimization
- Built Flask API with Gmail IMAP integration for real-time intent prediction and model serving, achieving 92% F1-score on test set through data augmentation via NLTK and back-translation

Portable AI: USB-Based Local LLM System

- Engineered fully offline LLM inference system with OpenAI-compatible API using llama.cpp and quantized GGUF models optimized for less than 8GB RAM environments, demonstrating model serving and optimization techniques
- Integrated SillyTavern frontend for prompt orchestration, memory handling, autonomous conversational workflows, and context window management for production-ready deployment